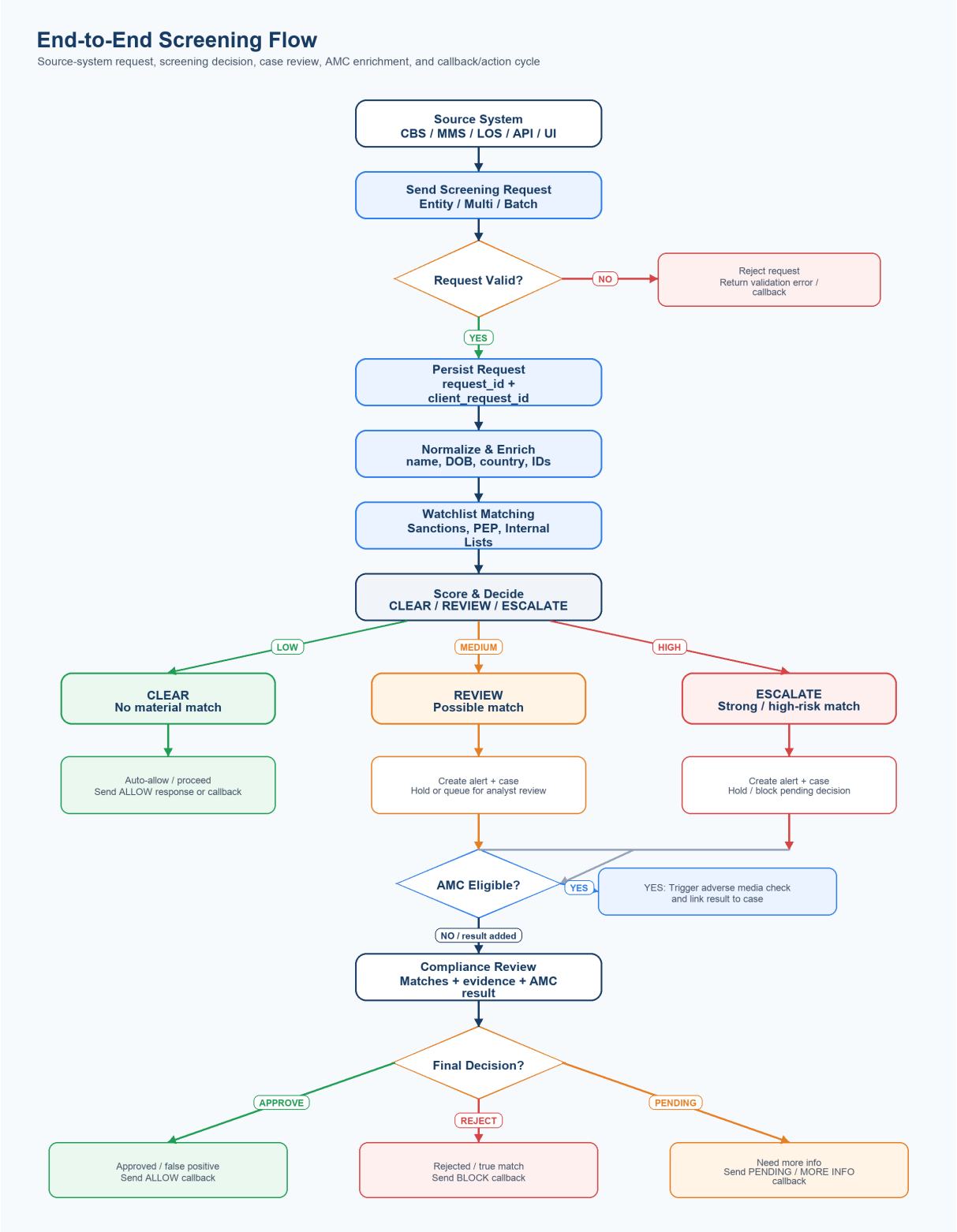
Screening Flow:



Screening CONFIG Requirements:

1. Source system code:
Sample: Merchant Onboarding System

2. Entity types they will send
Sample: ORGANIZATION, MERCHANT, LIMITED_LIABILITY_COMPANY

3. Relations under each entity
Sample:
ORGANIZATION -> CUSTOMER
ORGANIZATION -> DIRECTOR
ORGANIZATION -> AUTHORIZED_SIGNATORY
ORGANIZATION -> POC

4. Field mapping from their system to our API fields
SCREENING DATA POINTS REQUIRED:

1. Source System
Required / Optional: Required
What to provide: Upstream system sending the screening request, such as CBS, MMS, LOS, API, or UI.

2. Channel
Required / Optional: Required
What to provide: Business journey where screening is triggered, such as ONBOARDING, PERIODIC_REVIEW, or TRANSACTION.

3. Institution ID
Required / Optional: Required
What to provide: Client/institution code under which screening should be performed.

4. Entity Type
Required / Optional: Required
What to provide: Type of entity being screened, such as ORGANIZATION, MERCHANT, LIMITED_LIABILITY_COMPANY, NPO_NGO.

5. Entity Relation
Required / Optional: Required
What to provide: Role of the entity, such as CUSTOMER, DIRECTOR, POC, or AUTHORIZED_SIGNATORY.

6. External Entity ID
Required / Optional: Required for primary screening
What to provide: Unique entity/customer ID from the client's source system.

7. Primary Entity ID
Required / Optional: Required for secondary screening
What to provide: ID of the already-screened primary entity under which related entities are screened.

8. Relation Entity ID
Required / Optional: Optional
What to provide: Client-side ID for a related entity, such as director ID, POC ID, or signatory ID.

9. Name
Required / Optional: Required
What to provide: Legal name or full name of the person, merchant, organization, or related party.

10. Date of Birth
Required / Optional: Required for individuals/related persons
What to provide: Date of birth for individuals or related persons, preferably in YYYY-MM-DD format.

11. Country / Nationality
Required / Optional: Required
What to provide: Country, nationality, residence country, or incorporation country, preferably ISO alpha-2 code like IN, US, AE.

12. Address
Required / Optional: Optional but recommended
What to provide: Residential, registered, or operating address used as supporting screening evidence.

13. Gender
Required / Optional: Optional
What to provide: Gender of the individual, such as MALE or FEMALE, when available.

14. Entity Sub Type
Required / Optional: Optional
What to provide: More specific organization category, such as LLC, joint venture, NGO, or government body.

15. Identification Type 1
Required / Optional: Required
What to provide: Primary identification document type, such as PASSPORT, NATIONAL_ID, REGISTRATION_ID, TAX_ID, or OTHER.

16. Identification Number 1

Required / Optional: Required
What to provide: Primary identification document number/value.

17. Identification Type 2
Required / Optional: Optional
What to provide: Second identification document type, if available.

18. Identification Number 2
Required / Optional: Optional
What to provide: Second identification document number/value, if available.

19. Identification Type 3
Required / Optional: Optional
What to provide: Third identification document type, if available.

20. Identification Number 3
Required / Optional: Optional
What to provide: Third identification document number/value, if available.

21. Phone Number
Required / Optional: Optional
What to provide: Phone number associated with the entity or related person.

22. Email
Required / Optional: Optional
What to provide: Email address associated with the entity or related person.

23. Metadata
Required / Optional: Optional
What to provide: Additional client-side context such as batch ID, branch ID, user ID, workflow ID, or session ID.

24. Client Request ID
Required / Optional: Required for S2S/API
What to provide: Client-generated unique request reference used for status lookup, reconciliation, and idempotent tracking.

Sample Payload Data:

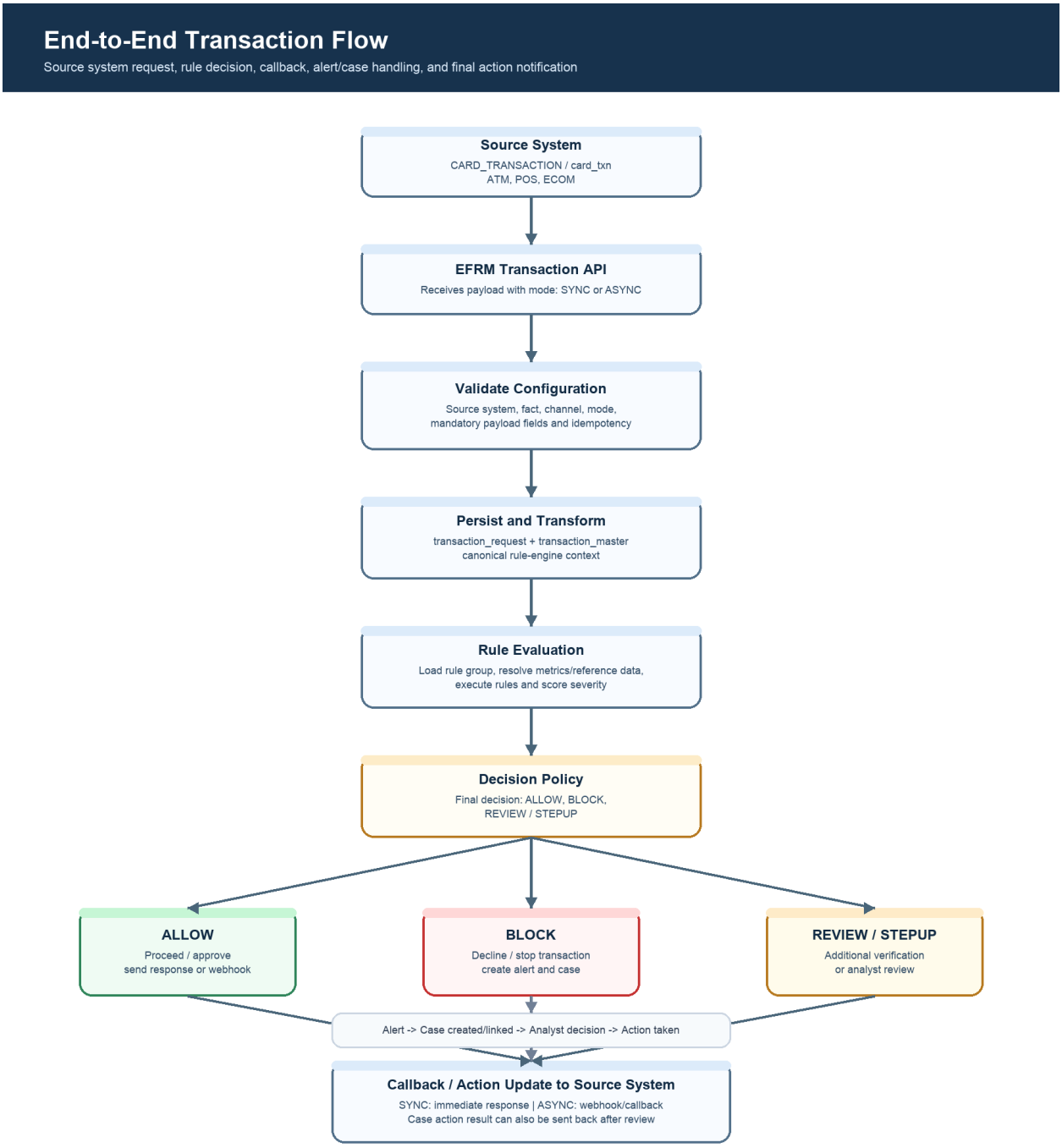
Request:


```
{
  "entity_type": "ORGANIZATION",
  "entity_relation": "CUSTOMER",
  "external_entity_id": "ORG-CLIENT-0001",
  "channel": "ONBOARDING",
  "source_system": "CBS",
  "institution_id": "KANJI",
  "client_request_id": "CBS-SCR-REQ-0001",
  "payload": {
    "name": "Example Trading LLC",
    "country": "AE",
    "address": "Office 1204, Business Bay, Dubai, United Arab
Emirates",
    "entity_sub_type": "LIMITED_LIABILITY_COMPANY",
    "id_type1": "REGISTRATION_ID",
    "id_value1": "AE-DXB-REG-123456",
    "id_type2": "TAX_ID",
    "id_value2": "TRN-100234567800003",
    "id_type3": "OTHER",
    "id_value3": "CHAMBER-CERT-998877",
    "phone_no": "+971501234567",
    "email": "compliance@exampletrading.ae",
    "metadata": {
      "branch_id": "DXB-001",
      "workflow_id": "ONB-2026-0001",
      "submitted_by": "relationship_manager_01"
    }
  }
}
```

Response:


```
{
  "request_id": "8d8c7a62-7a9f-4f51-8c0e-3f0d1d0c91a2",
  "client_request_id": "CBS-SCR-REQ-0001",
  "external_entity_id": "ORG-CLIENT-0001",
  "entity_type": "ORGANIZATION",
  "entity_relation": "CUSTOMER",
  "channel": "ONBOARDING",
  "source_system": "CBS",
  "institution_id": "KANJI",
  "decision": "ESCALATE",
  "risk_band": "HIGH",
  "overall_risk_score": 1000,
  "matched": true,
  "hit_count": 2,
  "alert_generated": true,
  "case_generated": true,
  "case_id": "CASE-20260810-000001",
  "message": "Potential screening match found. Compliance
review required.",
  "matches": [
    {
      "match_id": 101,
      "list_source": "OFAC",
      "matched_name": "EXAMPLE TRADING LLC",
      "match_score": 96.5,
      "risk_score": 1000,
      "risk_band": "HIGH",
      "match_type": "NAME",
      "matched_fields": {
        "name": true,
        "country": true,
        "identifier": true
      }
    }
  ],
  "recommended_action": "HOLD_FOR_REVIEW",
  "screened_at": "2026-08-10T14:30:10Z"
}
```

Transaction Flow:



Transaction flow from source request to decision, callback, case handling, and final action update.

2)TRANSACTION CONFIG REQUIREMENTS:

1. **Source systems**
   Example :`CARD_TRANSACTION`, `DEVICE_EVENT`, `FOREX_TRANSACTION`

2. Facts

Example: `card_txn`, `sign3_device_data`, `kanji_device_data`, `bugs_device_data`, `forex_txn`

3. Channels

Example: For `CARD_TRANSACTION / card_txn`: `ATM`, `POS`, `ECOM`

4. Mode

`SYNC`, `ASYNC`

5. Incoming payload fields for `card_txn`:

`institution_id`,
`source_system`, `mode`,
`channel`,
`source_txn_id`,
`request_id`,
`txnreqtrandate`,
`pan`,
`processing_code`,
`amount`,
`stan`,
`local_time`,
`local_date`,
`mcc`,
`country_code`,
`pos_entry_mode`,
`pos_condition_code`,
`pin_capture_code`,
`acquiring_institution_id`,
`forwarding_institution_id`,
`track2_data`,
`rrn`,
`authorization_id`,
`response_code`,
`terminal_id`,
`customer_id`,
`merchant_name_location`,
`currency_code`

6. SYNC SAMPLE PAYLOAD:


```
{
  "payload": {
    "institution_id": "KANJI",
    "source_system": "CARD_TRANSACTION",
    "mode": "SYNC",
    "channel": “ECOM",
    "source_txn_id": "SRC-SYNC-000001",
    "request_id": "REQ-SYNC-000001",
    "txnreqtrandate": "260810143000",
    "pan": "4111111111111111",
    "processing_code": "010000",
    "amount": 3200,
    "stan": "123456",
    "local_time": "143000",
    "local_date": "0810",
    "mcc": "5411",
    "country_code": "356",
    "pos_entry_mode": "051",
    "pos_condition_code": "00",
    "pin_capture_code": "12",
    "acquiring_institution_id": "ACQ001",
    "forwarding_institution_id": "FWD001",
    "track2_data": "4111111111111111=27081230000012345678",
    "rrn": "626081000001",
    "authorization_id": "",
    "response_code": "",
    "terminal_id": "TERM001",
    "customer_id": "CUST000001",
    "merchant_name_location": "TEST MERCHANT CITY IN",
    "currency_code": "356"
  }
}
```

7. ASYNC SAMPLE PAYLOAD:


```
{
  "payload": {
    "institution_id": "KANJI",
    "source_system": "CARD_TRANSACTION",
    "mode": "ASYNC",
    "channel": "POS",
    "source_txn_id": "SRC-SYNC-000001",
    "request_id": "REQ-ASYNC-000001",
    "txnreqtrandate": "260810143000",
    "pan": "4111111111111111",
    "processing_code": "010000",
    "amount": 3200,
    "stan": "123456",
    "local_time": "143000",
    "local_date": "0810",
    "mcc": "5411",
    "country_code": "356",
    "pos_entry_mode": "051",
    "pos_condition_code": "00",
    "pin_capture_code": "12",
    "acquiring_institution_id": "ACQ001",
    "forwarding_institution_id": "FWD001",
    "track2_data": "4111111111111111=27081230000012345678",
    "rrn": "626081000001",
    "authorization_id": "AUTH001",
    "response_code": "00",
    "terminal_id": "TERM001",
    "customer_id": "CUST000001",
    "merchant_name_location": "TEST MERCHANT CITY IN",
    "currency_code": "356"
  }
}
```

SYNC Response Example

For `SYNC` mode, EFRM returns the decision immediately in the API response.


```
{
  "request_id": "REQ-SYNC-000001",
  "source_txn_id": "SRC-SYNC-000001",
  "institution_id": "KANJI",
  "source_system": "CARD_TRANSACTION",
  "fact": "card_txn",
  "mode": "SYNC",
  "channel": "ECOM",
  "decision": "ALLOW",
  "risk_score": 120,
  "risk_severity": "LOW",
  "policy_code": "POLICY_1",
  "matched_rule_count": 0,
  "alert_generated": false,
  "signals": [],
  "message": "Transaction allowed"
}
```


BLOCK Response Example


```
{

  "request_id": "REQ-SYNC-000002",
  "source_txn_id": "SRC-SYNC-000002",
  "institution_id": "KANJI",
  "source_system": "CARD_TRANSACTION",
  "fact": "card_txn",
  "mode": "SYNC",
  "channel": "POS",
  "decision": "BLOCK",
  "risk_score": 850,
  "risk_severity": "CRITICAL",
  "policy_code": "POLICY_1",
  "matched_rule_count": 2,
  "alert_generated": true,
  "signals": [
    {
      "rule_code": "POS_HIGH_VALUE_TXN",
      "signal_code": "POS_HIGH_VALUE_TXN",
      "severity": "CRITICAL",
      "score": 850,
      "description": "High value POS transaction detected"
```

```
    }
  ],
  "message": "Transaction blocked due to high risk"
}
```

ASYNC Response Example

For `ASYNC` mode, the API can accept the request first and later notify the source system through webhook/callback.

```
{
  "request_id": "REQ-ASYNC-000001",
  "source_txn_id": "SRC-SYNC-000001",
  "institution_id": "KANJI",
  "source_system": "CARD_TRANSACTION",
  "fact": "card_txn",
  "mode": "ASYNC",
  "status": "ACCEPTED",
  "message": "Transaction received for asynchronous risk
evaluation"
```

8. Decision Response Format

After rule execution, EFRM returns or sends a decision result. The same decision structure can be used for both `SYNC` response and `ASYNC` webhook/callback.

Decision Values

Decisio	Meaning	Expected Client Action
`ALLOW`	Transaction is low risk and can proceed	Approve / continue transaction
`BLOCK`	Transaction is high risk and should not proceed	Decline / stop transaction
`REVIEW`	Transaction requires analyst review	Hold, review, or process as per client policy
`STEPUP`	Transaction requires additional verification	Trigger OTP, PIN, biometric, or challenge flow

```
}
```

9. Action and Callback/Webhook Format

Actions define what EFRM or the client should do after the decision is generated.

Action Types

Action Type	When Used	Description
`RETURN_SYNC_RESPONSE`	`SYNC` mode	Return decision immediately in transaction API response
`SEND_WEBHOOK`	`ASYNC` mode or post-case update	Send decision/action result to client callback URL
`CREATE_ALERT`	Risky transaction	Create alert in EFRM alert queue
`CREATE_CASE`	Review/block transaction	Create or link alert to a case
`STEPUP_REQUIRED`	Medium-risk transaction	Ask client system to perform OTP/PIN/biometric/challenge
`BLOCK_TRANSACTION`	High/critical risk transaction	Ask client/source system to decline or stop transaction
`MARK_FOR_REVIEW`	Review-required transaction	Mark transaction for analyst or compliance review
`LOG_ONLY`	Low risk or shadow mode	Store result without alert/case/action

Decision to Action Mapping

Mode	Decision	EFRM Action	Source System Action
`SYNC`	`ALLOW`	Return allow response	Proceed with transaction
`SYNC`	`BLOCK`	Return block response, create alert/case if configured	Decline transaction
`SYNC`	`REVIEW`	Return review response, create alert/case	Hold or process as per client policy
`SYNC`	`STEPUP`	Return step-up response	Trigger OTP/PIN/challenge
`ASYNC`	`ALLOW`	Store result, optionally send webhook	Record safe/low-risk result
`ASYNC`	`BLOCK`	Create alert/case and send webhook	Take post-transaction action as per policy
`ASYNC`	`REVIEW`	Create alert/case and send webhook	Review transaction/customer

ASYNC	STEP UP	Create alert or send webhook	Apply follow-up verification if applicable

Webhook / Callback Request Format

EFRM can send the final decision to the source system using the client-provided callback URL.


```
{
  "event_type": "TRANSACTION_DECISION",
  "event_id": "EVT-20260810-000001",
  "event_time": "2026-08-10T14:30:10Z",
  "institution_id": "KANJI",
  "source_system": "CARD_TRANSACTION",
  "fact": "card_txn",
  "mode": "ASYNC",
  "channel": "POS",
  "request_id": "REQ-ASYNC-000001",
  "source_txn_id": "SRC-SYNC-000001",
  "decision": "BLOCK",
  "risk_score": 850,
  "risk_severity": "CRITICAL",
  "policy_code": "POLICY_1",
  "matched_rule_count": 2,
  "alert_generated": true,
  "case_created": true,
  "case_id": "CASE-20260810-000001",
  "signals": [
    {
      "rule_code": "POS_HIGH_VALUE_TXN",
      "signal_code": "POS_HIGH_VALUE_TXN",
      "severity": "CRITICAL",
      "score": 850,
      "description": "High value POS transaction detected"
    }
  ],
  "recommended_action": "BLOCK_TRANSACTION",
  "message": "Transaction blocked due to high risk"
}
```


Webhook / Callback Response Expected from Source System

The source system should acknowledge that it received the webhook.


```
{
  "event_id": "EVT-20260810-000001",
  "source_txn_id": "SRC-SYNC-000001",
  "status": "RECEIVED",
```

```
  "received_at": "2026-08-10T14:30:12Z",
  "message": "Decision received successfully"
}
```

Case Action Callback Format

If a case is created and later closed after analyst review, EFRM can send the final case action/update to the source system.

```
{
  "event_type": "CASE_ACTION_UPDATE",
  "event_id": "EVT-20260810-000002",
  "event_time": "2026-08-10T15:10:00Z",
  "institution_id": "KANJI",
  "source_system": "CARD_TRANSACTION",
  "source_txn_id": "SRC-SYNC-000001",
  "case_id": "CASE-20260810-000001",
  "case_status": "CLOSED",
  "final_action": "CONFIRMED_FRAUD",
  "decision": "BLOCK",
  "remarks": "Analyst confirmed suspicious transaction
pattern",
  "message": "Case closed and final action recorded"
}
```

Client Configuration Needed

Configuration	What Client Should Provide
Supported transaction modes	`SYNC`, `ASYNC`, or `BOTH`
Supported channels	Example: `ATM`, `POS`, `ECOM`
Callback/webhook URL	Endpoint where EFRM should send async decisions and case updates
Webhook authentication method	API key, bearer token, mTLS, or signed payload
Decision mapping	How client maps `ALLOW`, `BLOCK`, `REVIEW`, `STEPUP` internally
Response code mapping	Client-specific codes for allow/block/review/step-up
Step-up method	OTP, PIN, biometric, device challenge, or other method
Case/alert preference	Whether EFRM should auto-create alerts/cases for block/review decisions
Retry policy	Number of webhook retries and retry interval
Timeout policy	Maximum wait time for webhook acknowledgement

Decision Callback Sample:


```
{
  "event_type": "TRANSACTION_DECISION",
  "event_id": "EVT-20260810-000001",
  "event_time": "2026-08-10T14:30:10Z",
  "institution_id": "KANJI",
  "source_system": "CARD_TRANSACTION",
  "fact": "card_txn",
  "mode": "ASYNC",
  "channel": "POS",
  "request_id": "REQ-ASYNC-000001",
  "source_txn_id": "SRC-SYNC-000001",
  "decision": "BLOCK",
  "risk_score": 850,
  "risk_severity": "CRITICAL",
  "policy_code": "POLICY_1",
  "matched_rule_count": 2,
  "alert_generated": true,
  "case_created": true,
  "case_id": "CASE-20260810-000001",
  "signals": [
    {
      "rule_code": "POS_HIGH_VALUE_TXN",
```

```
      "signal_code": "POS_HIGH_VALUE_TXN",
      "severity": "CRITICAL",
      "score": 850,
      "description": "High value POS transaction detected"
    }
  ],
  "recommended_action": "BLOCK_TRANSACTION",
  "message": "Transaction blocked due to high risk"
}
```